EICOSANOIDS

(prostaglandins, thromboxanes, leukotrienes)

Eicosanoids

- Major classes of eicosanoids.
- Precursors of eicosanoids.
- Major pathways of eicosanoid synthesis (cyclooxygenase and lipoxygenase).
- Important functions of eicosanoids.
- Important inhibitors of eicosanoid synthesis

Eicosaniods

- Derived from 20-crabon polyunsaturated fatty acids
- Paracrine or autocrine messengers molecules
- Short half-lives (10 secs – 5 mins) so that functions are usually limited to actions on nearby cells.
- Bind to specific cell surface G-protein coupled receptors, and generally increase cAMP levels. May also bind to nuclear receptors and alter gene transcription.
- Wide variety of functions

Major Classes of Eicosanoids

- Prostaglandins
- Thromboxanes
- Prostacyclins
- Leukotrienes
- HETES

Effects of Eicosaniods

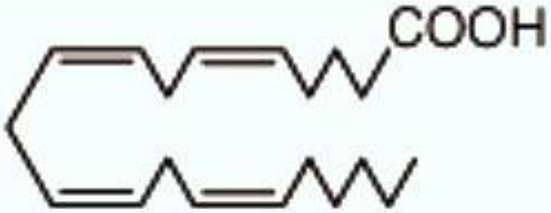
- Induction of inflammation
- Mediation of pain signals
- Induction of fever
- Smooth muscle contraction (including uterus)
- Smooth muscle relaxation
- Protection of stomach lining
- Simulation of platelet aggregation
- Inhibition of platelet aggregation
- Sodium and water retention

Precursors of Eicosanoids

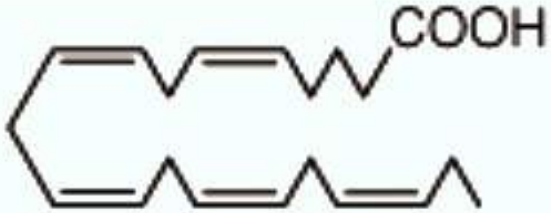
- Arachidonic acid (ω6)
- Eicosatrienoic acid (γ-linolenic acid, ω6)
- Eicosapentaenoic Acid (ω3)



Eicosatrienoic acid



Arachidonic acid



Eicosapentaenoic acid

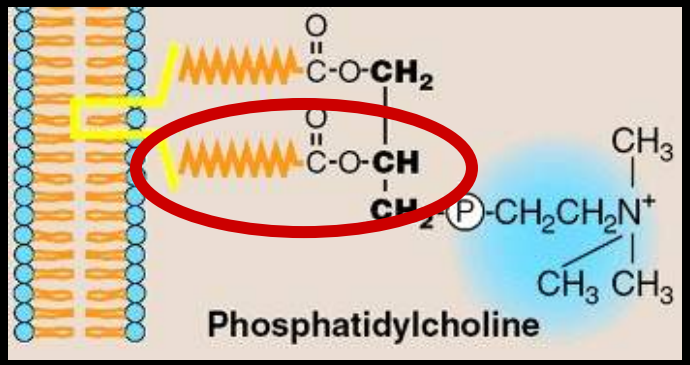
Dietary Linoleic Acid (C18: $\Delta^{9,12}$) (from plant oils)

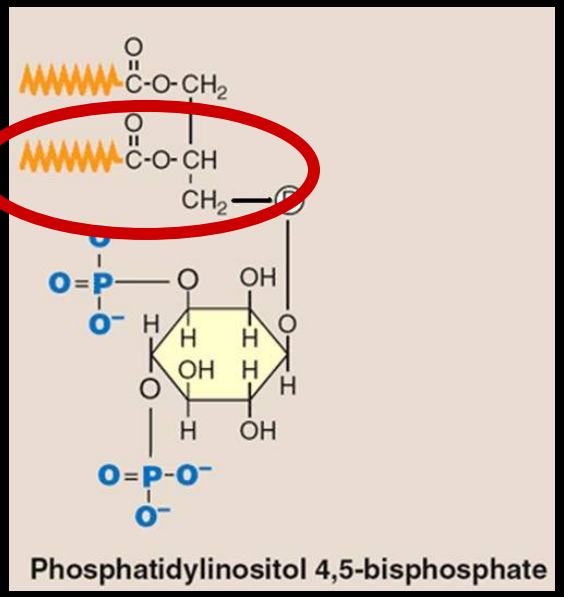
Elongase

Desaturase

Arachidonic Acid (C20: $\Delta^{5, 8, 11,}$

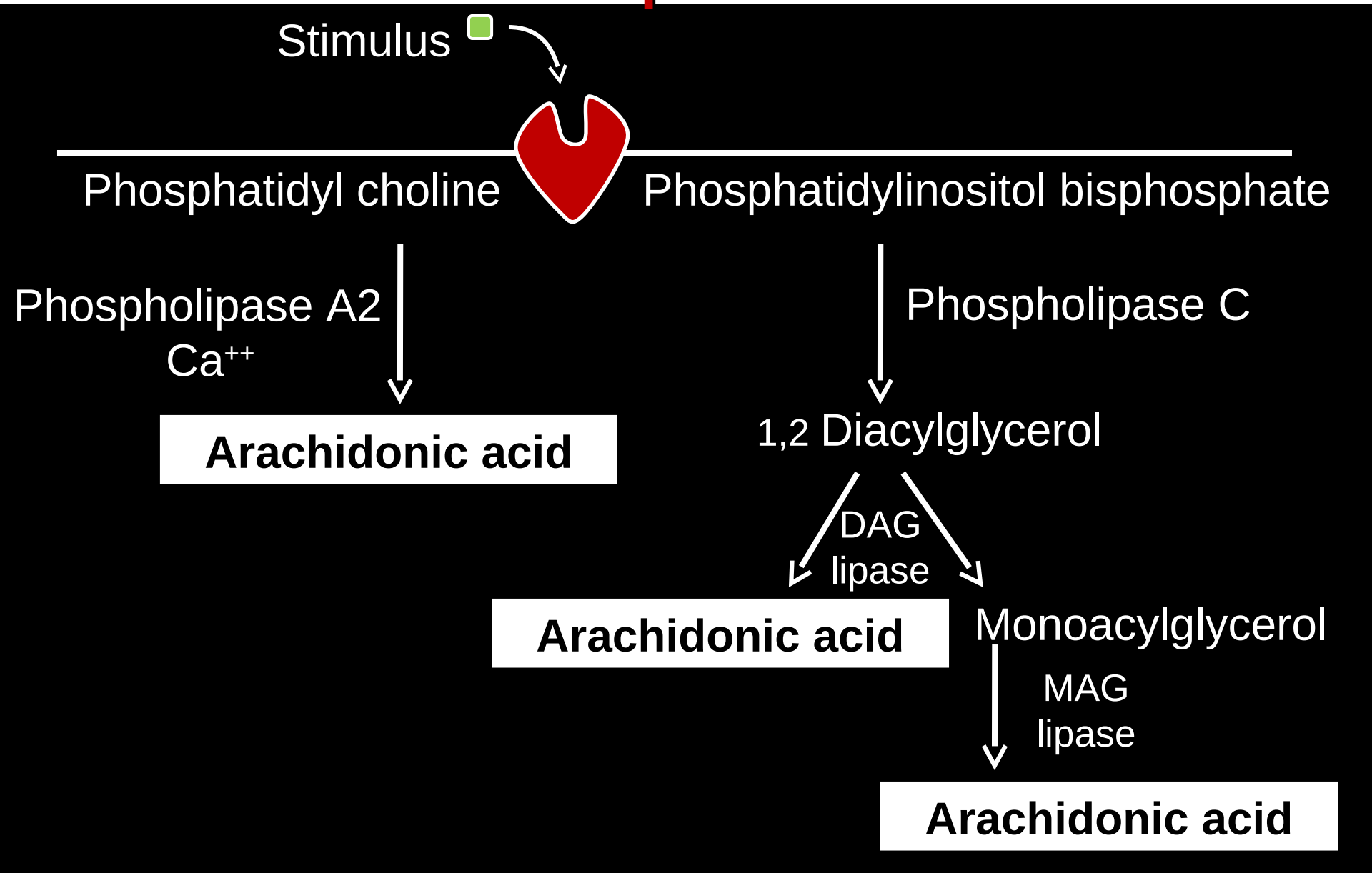
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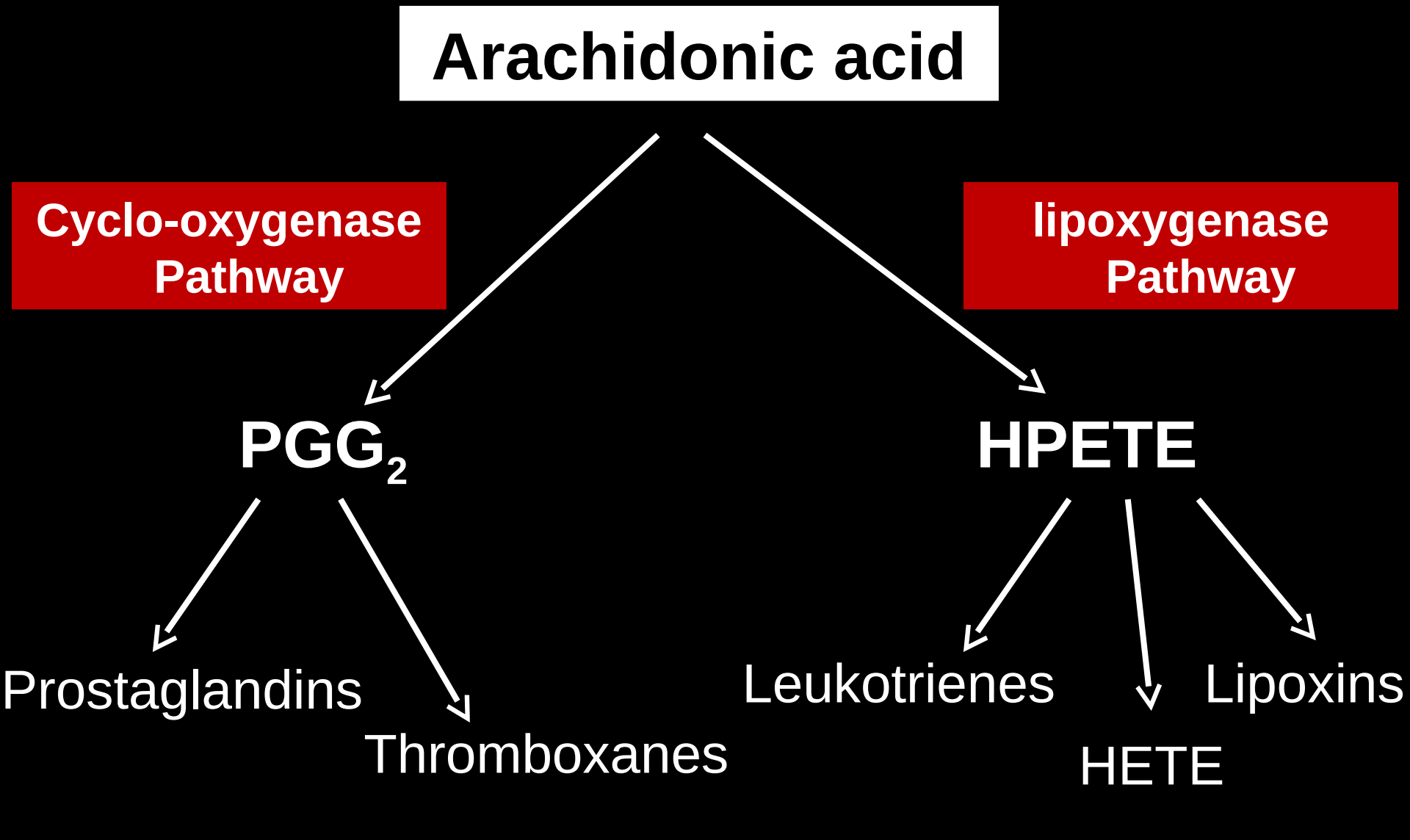


Membrane Phospholipids

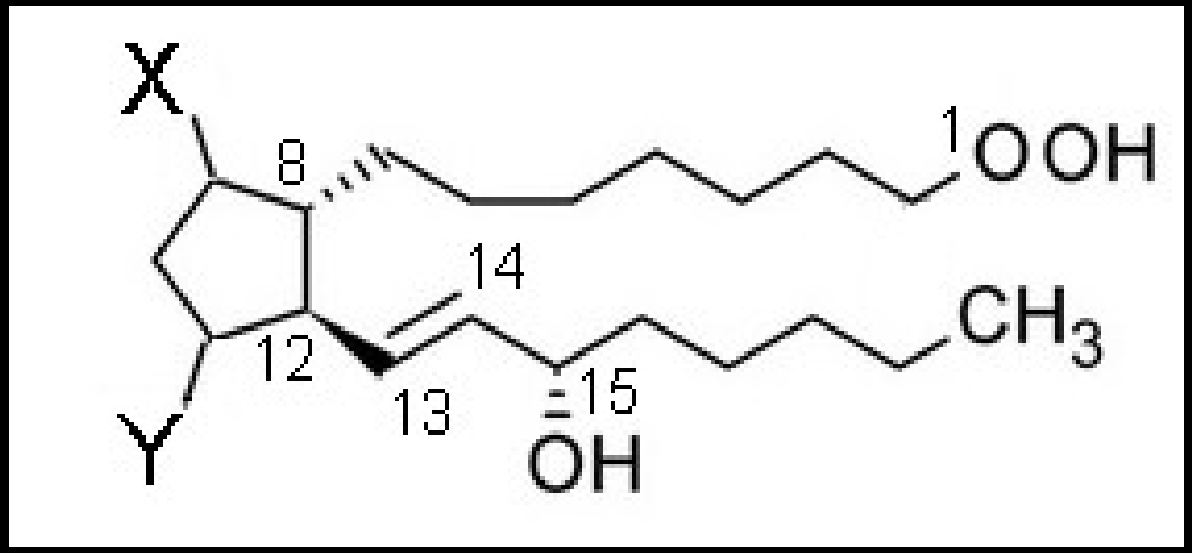
Arachidonic acid release from membrane lipids



Pathways for Arachidonic Acid Metabolism



Prostaglandins – Structural Features



PGA, PGD, PGE, PGF, PGG, PGH, PGI

Depending on the functional groups present at X and Y

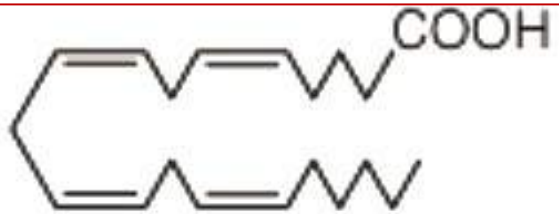
PGF $_{1, 2}$ or $_{3}$

Depending on the number of double bonds present in the linear hydrocarbon chain

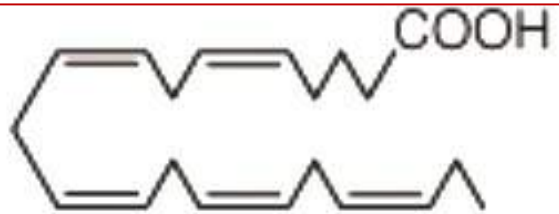
PGF $_{1, 2}$ or $_{3}$



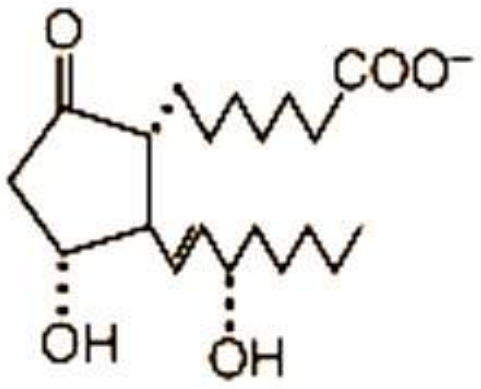
Eicosatrienoic acid



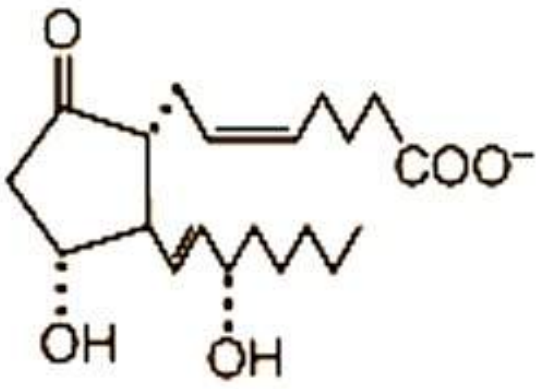
Arachidonic acid



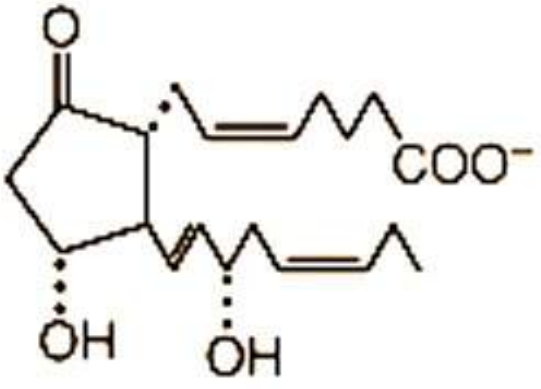
Eicosapentaenoic acid



PGE_1

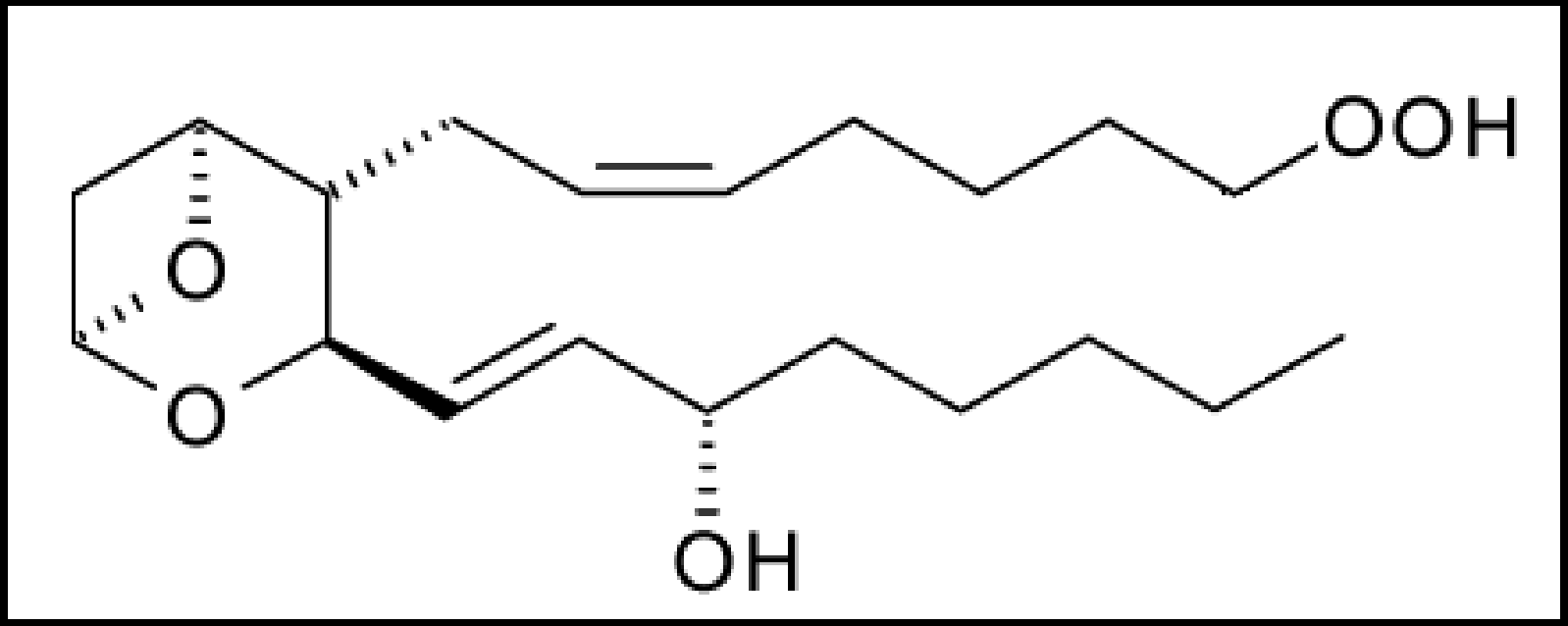


PGE_2



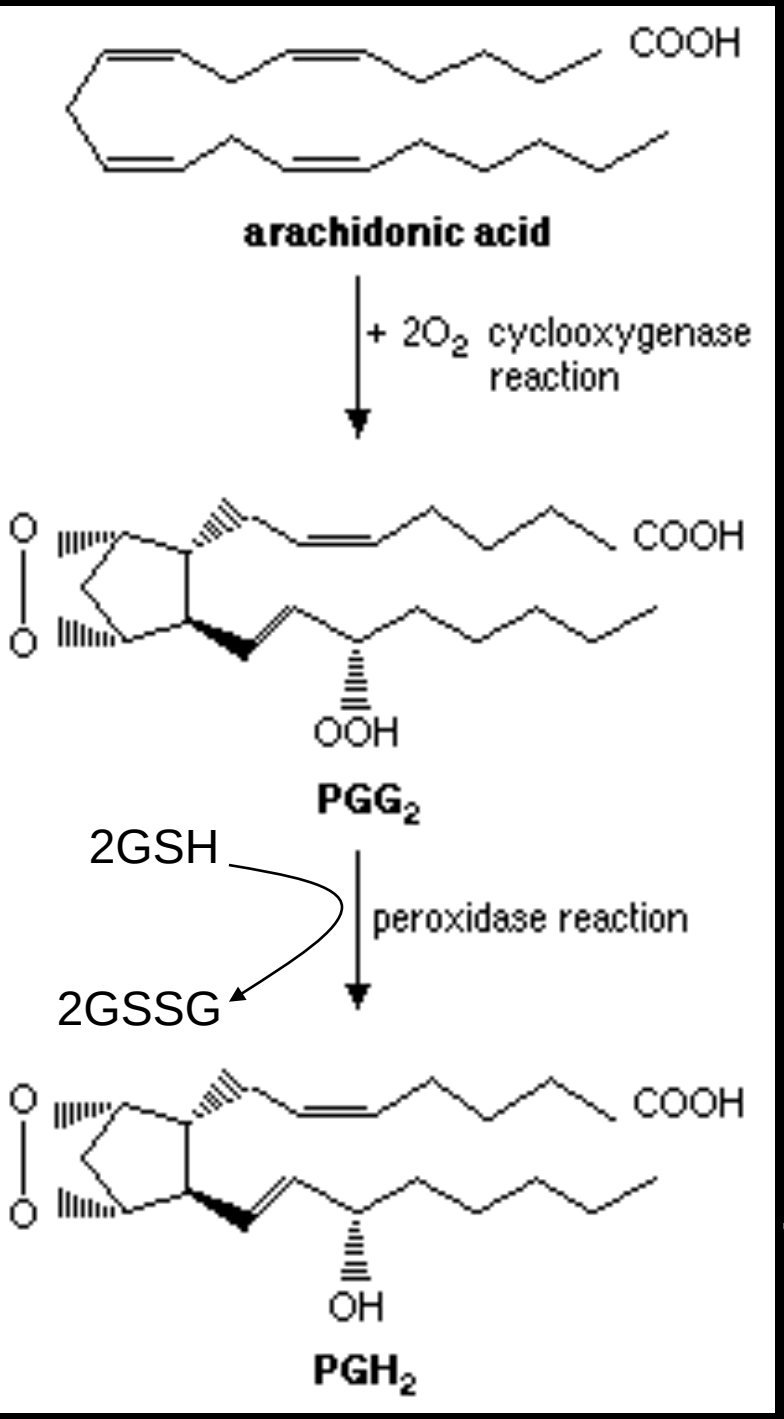
PGE_3

Thromboxane A_2 (TXA_2) - structure



CYCLO-OXYGENASE PATHWAY

PG and TX synthesis



Tissue specific

PGD synthase → PGD_2

PGE synthase → PGE_2

PGE_2 → (PGE 9-keto reductase) → PGF_{2a}

PGI synthase → PGI_2

TXA synthase → TXA_2

Some Functions of Prostaglandins

PGI2, PGE2, PGD2

- ↑ Vasodilation, cAMP
- ↓ Platelet and leukocyte aggregation, IL1 and IL2, T-cell proliferation, lymphocyte migration

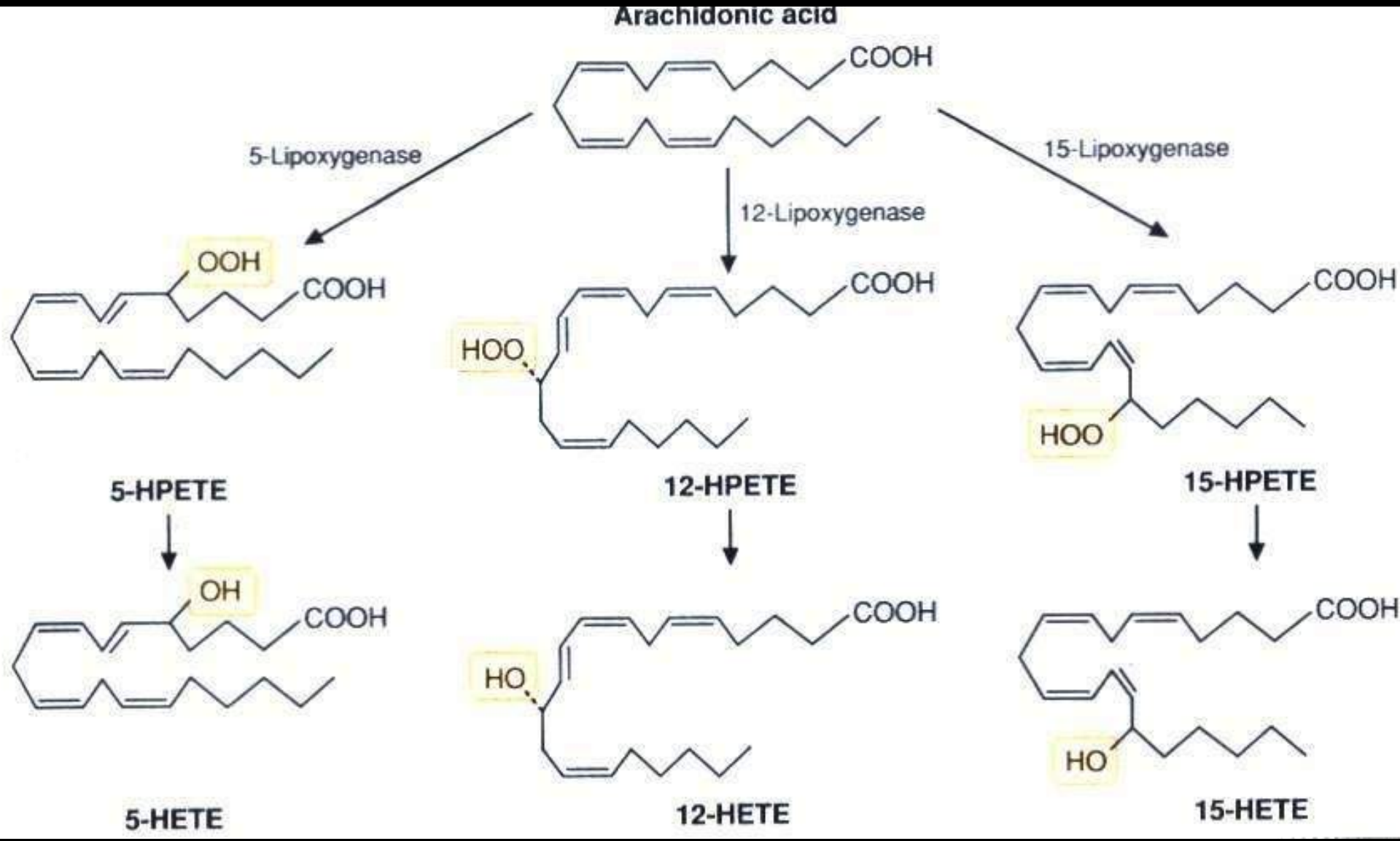
PGF2α

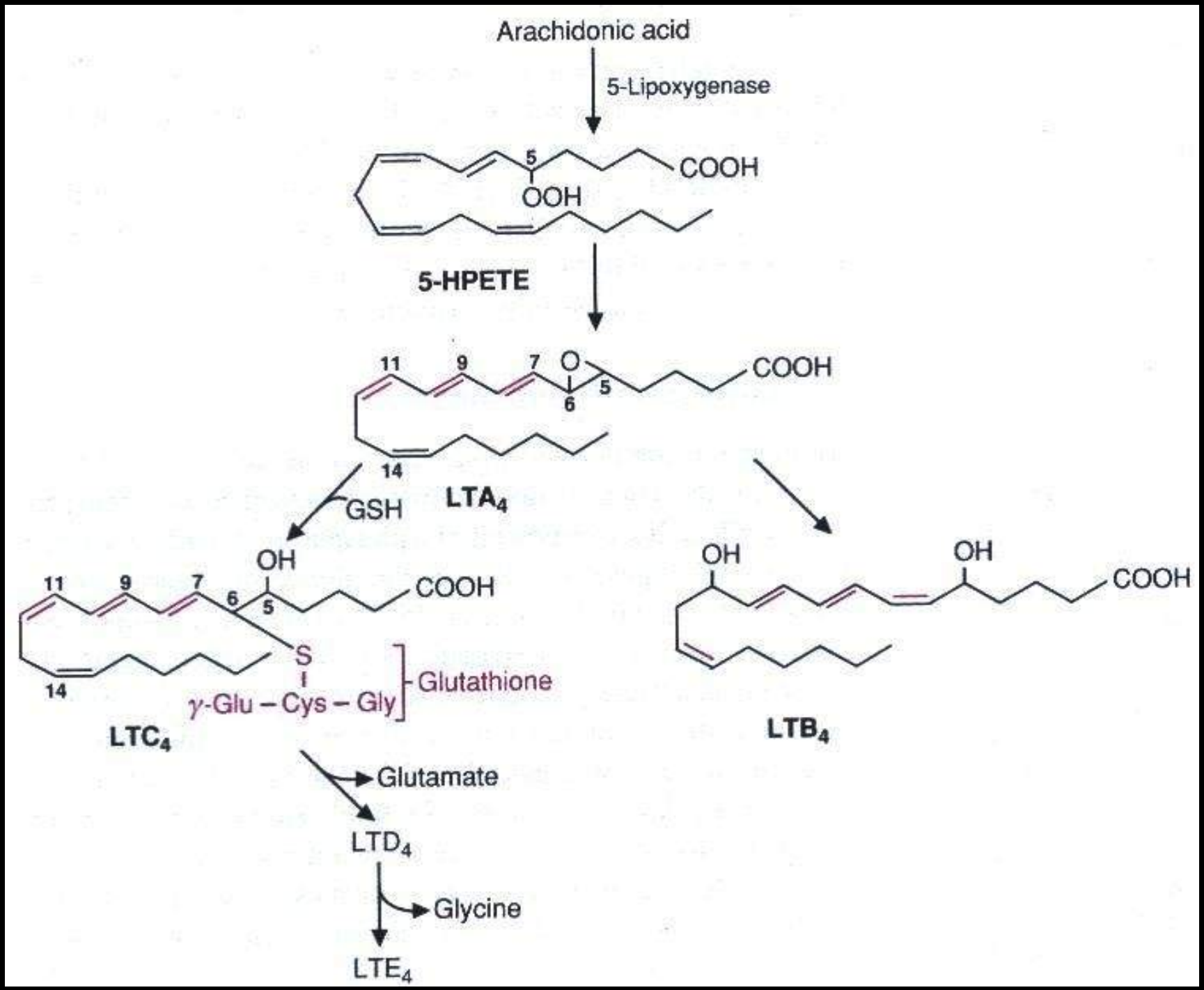
- ↑ Vasoconstriction, Bronchoconstriction, smooth muscle contraction

TXA2

- ↑ Vasoconstriction, Platelet aggregation, lymphocyte proliferation, bronchoconstriction

Lipoxygenase pathway




Arachidonic acid
5-Lipoxygenase
5
COOH
OOH
5-HPETE
11
9
7
O
5
6
14
COOH
LTA4
GSH
OH
11
9
7
6
5
COOH
14
S
γ-Glu – Cys – Gly
Glutathione
LTC4
OH
OH
COOH
LTB4
Glutamate
LTD4
Glycine
LTE4

Some Functions of Leukotrienes

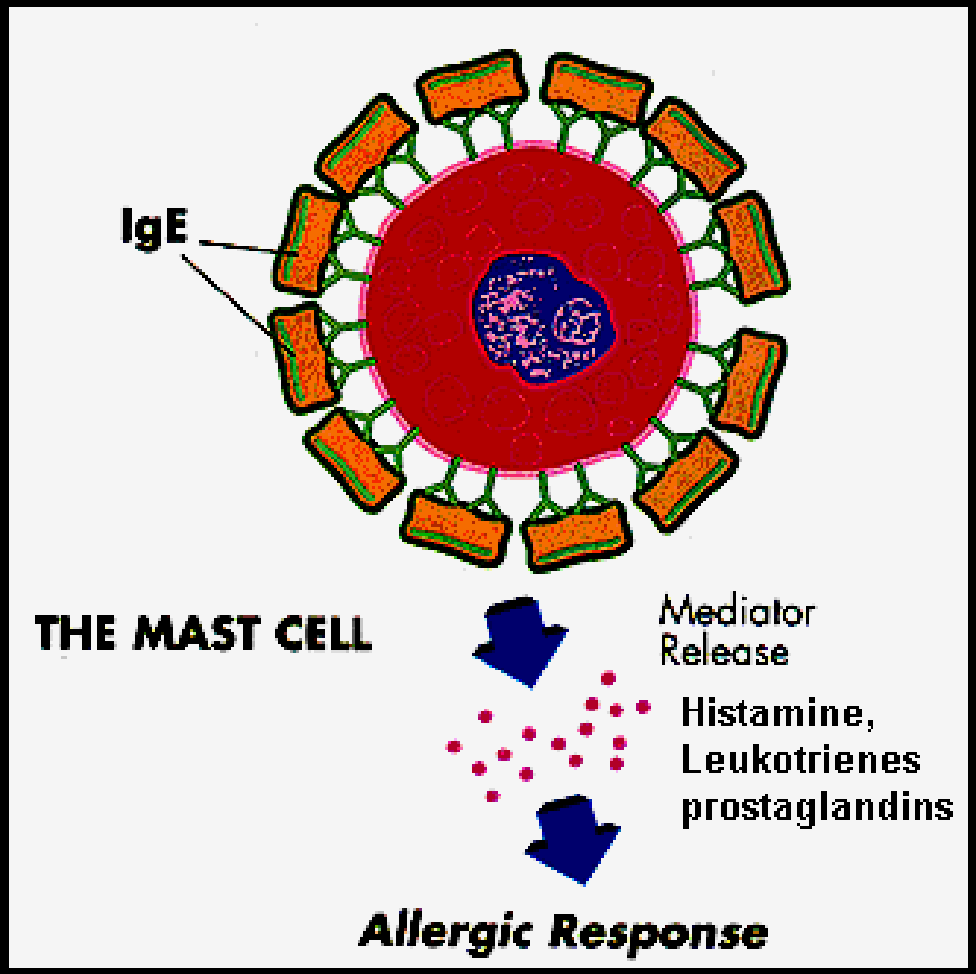
LTB4

- ↑ Vascular permeability, T-cell proliferation, leukocyte aggregation, IL -1, IL-2, IFN-γ

LTC4 and LTD4

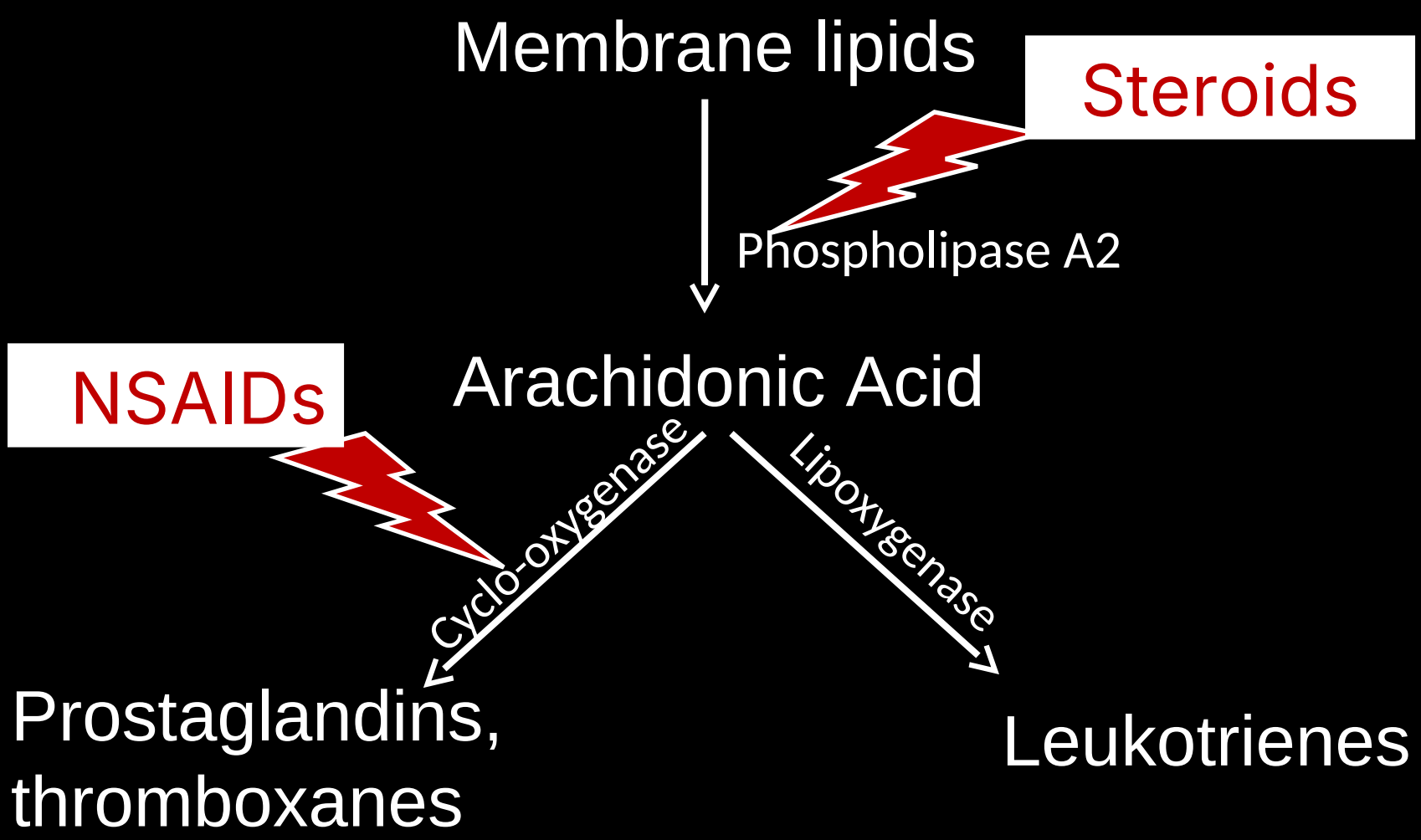
- ↑ Bronchoconstriction, Vascular permeability, IFN-γ

Leukotrienes and allergies

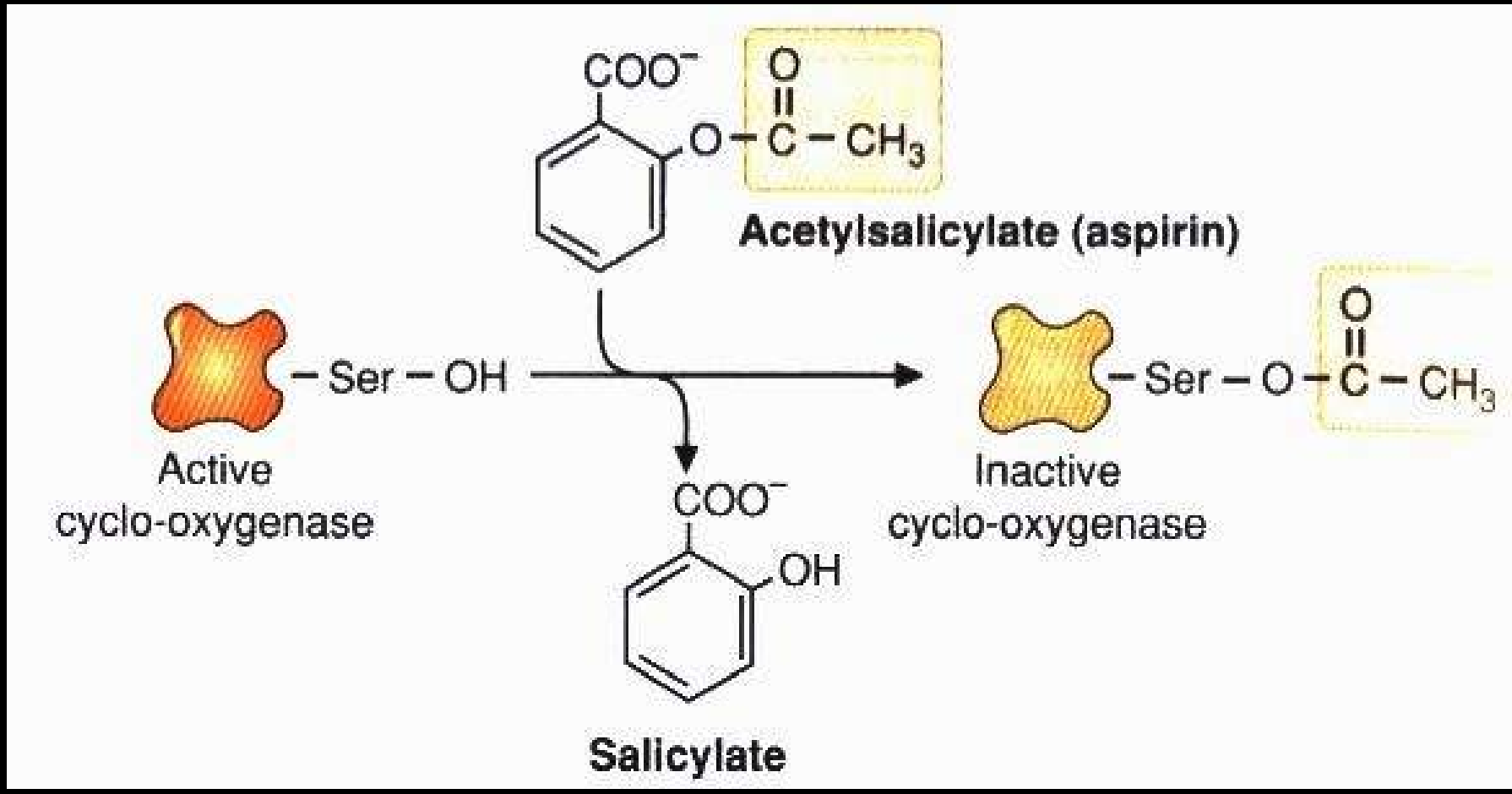


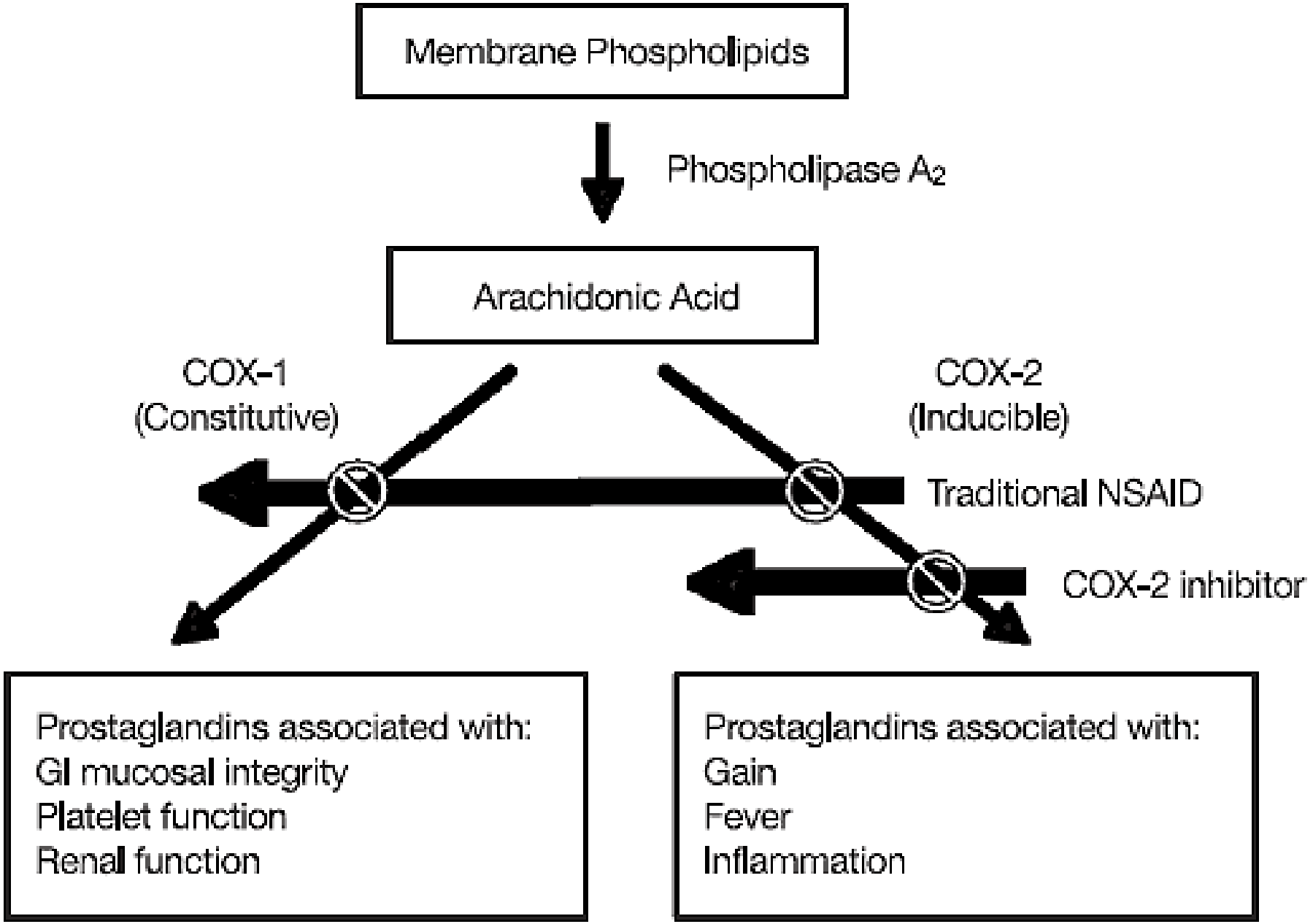
- Leukotrienes are a hundred times more potent than histamine
- Histamine provided a rapid response to an allergen
- In the later stages leukotrienes are principally responsible for inflammation, smooth muscle constriction, constriction of the airways and mucous secretion form mucosal epithelium

Anti inflammatory Drugs inhibit Eicosanoid Synthesis



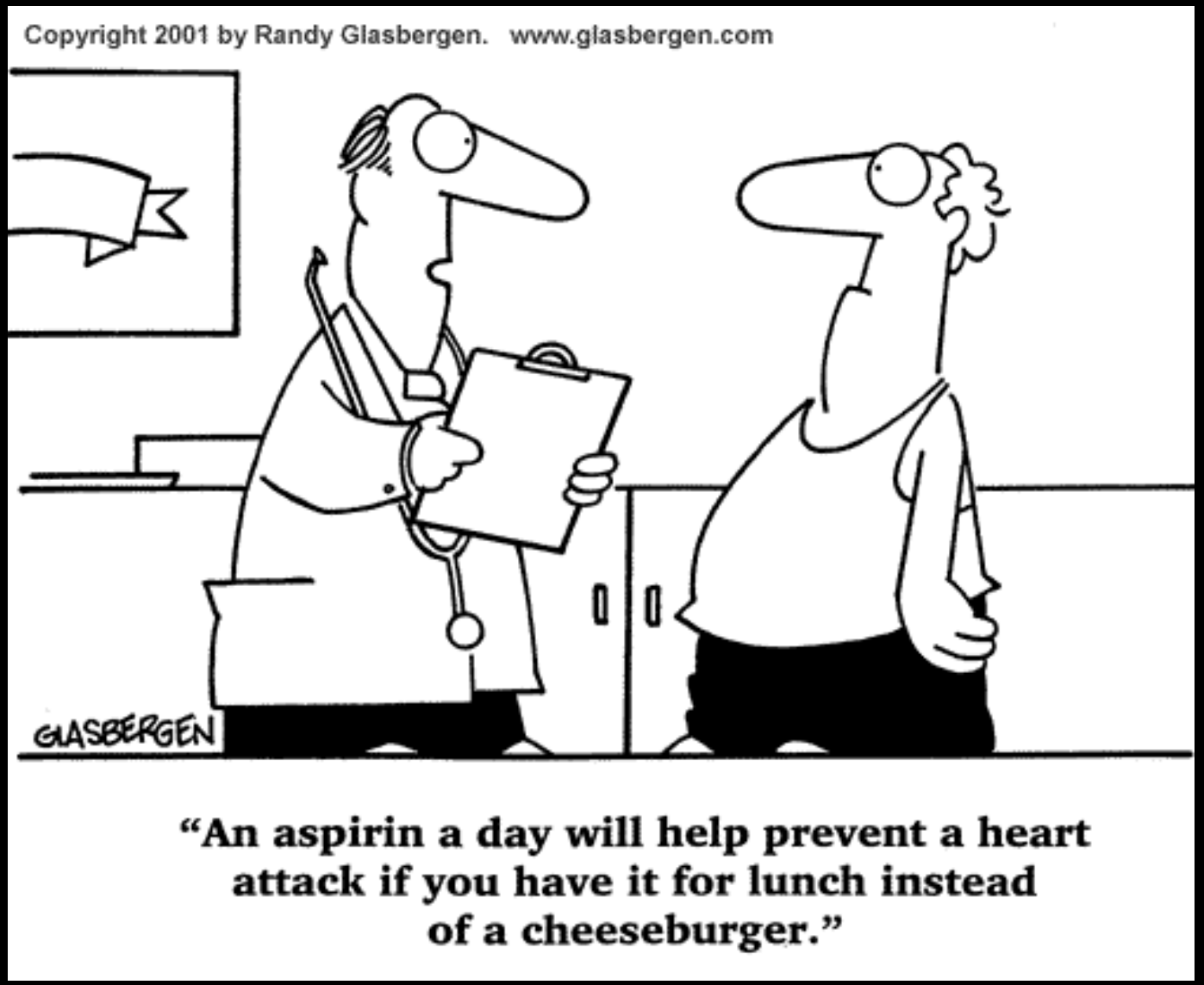
Mechanism of Aspirin Action





NSAID = nonsteroidal anti-inflammatory drug; COX = cyclooxygenase; GI = gastrointestinal.

Aspirin and cardiovascular disease



Omega-6/omega-3 fatty acid balance

- ω6 and ω3 are not interconvertible in humans (mammals).
- Diets rich in ω3 fatty acids result in high content of these fatty acids in membrane phospholipids

Recommended ratio: 1-4: 1 (ω6 : ω3)

Typical western diet: 14-25: 1 (ω6 : ω3)

Omega-6/omega-3 fatty acid balance

A diet rich in omega-6 FAs shifts the physiological state to one that is proinflammatory, prothrombotic and proaggregatory… leading to heart disease in susceptible individuals

Table 5. Ethnic Differences in Fatty Acid Concentrations in Thrombocyte Phospholipids and Percentage of All Deaths from Cardiovascular Disease[a]

	Europe and United States (%)	Japan (%)	Greenland Eskimos (%)
Arachidonic acid (20:4ω6)	26	21	8.3
Eicosapentaenoic acid (20:5ω3)	0.5	1.6	8.0
Ratio of ω6/ω3	50	12	1
Mortality from cardiovascular disease	45	12	7

Thank you!